

Reliability Study Report

20% Cement Foam Specimens — Reusability Assessment

March 2026

Background & Motivation

This study assesses the reusability of 20% cement foam specimens by re-testing the same specimens across multiple days. The original run (March 18) established a baseline evaporation rate. The same specimens were then re-tested on Day 1 and Day 2 following the reliability protocol below. The goal is to determine whether foam performance degrades, stabilises, or recovers with repeated use cycles — a key factor in evaluating the material for practical deployment.

Reliability Test Procedure

Per the instructions of Dr. Boymelgreen, the following protocol was followed for all reliability runs. Note: insufficient samples were available to compare dry re-wetted vs. continuously wetted foam, so all specimens were kept continuously wetted between runs.

1. Each foam specimen was placed individually in a petri dish with deionised (DI) water — enough to maintain moisture without waterlogging — and soaked for a minimum of 1 hour to ensure full wetting prior to testing.
2. Using a fresh piece of weighing paper on a tared scale, the wet foam specimen was weighed and the mass recorded in the lab notebook.
3. The scale was cleared and re-tared. The PLA evaporation disk was placed on the balance and 1 mL of DI water was added via dropper. The combined weight was recorded.
4. The wetted foam specimen was placed into the evaporation disk containing the DI water, and the total combined weight was recorded.
5. The assembly was transferred to the solar simulator and the 3600s data acquisition was started as per the standard procedure.

Mass Loss Rate Formula:

$$\dot{m} [\text{kg/m}^2/\text{hr}] = \frac{\text{Slope} [\text{g/s}]}{1000} \times \frac{(100 \text{ cm/m})^2}{9 \text{ cm}^2} \times 3600 \text{ s/hr}$$

Day 1 Re-test — Same Specimens (Baseline Comparison)

Sample	Initial (g)	Final (g)	Δm (g)	Loss (kg/m ² /hr)
Original S1	12.9928	11.3894	1.6034	1.8646
Original S2	12.9603	11.3840	1.5763	1.8274
Original S3	12.9476	11.4519	1.4957	1.7352
Original Avg	12.9669	—	1.5585	1.8091
Day 1 S1	12.8536	11.5895	1.2641	1.4670
Day 1 S2	12.4964	11.1064	1.3900	1.6101
Day 1 S3	12.4886	11.1014	1.3872	1.5953
Day 1 Avg	12.6129	—	1.3471	1.5575
Change vs. Original	−0.354 g	—	−0.211 g	−13.9%

Observations:

- All three Day 1 samples return strong R^2 values (S1: 0.9987, S2: 0.9992, S3: 0.9992), confirming the foam maintains clean linear evaporation behaviour after one full use cycle.
- **The average mass loss rate dropped 13.9%** (1.8091 $\rightarrow$ 1.5575 kg/m²/hr). With three samples confirming this trend, it is not attributable to a single outlier.
- S2 and S3 are very consistent (1.6101 vs. 1.5953 kg/m²/hr, only 0.9% apart). S1 sits lower at 1.4670, possibly reflecting environmental conditions at the time of testing.
- **Initial weights declined across all three** (avg 12.97 g $\rightarrow$ 12.61 g), indicating the foam holds less water after one use cycle even when kept wet between runs.
- **Δm declined** from 1.559 g to 1.347 g on average, consistent with reduced surface water availability.
- The rolling rate overlay confirms the Day 1 samples track consistently below the originals throughout the full run — the reduction is systematic, not an artefact.
- **Interpretation:** a 13.9% reduction after one cycle is meaningful. Some portion is likely environmental (ambient conditions differed between test days), but the consistent initial weight decline across all three specimens points to genuine foam behaviour change, likely a reduction in effective pore volume. Further cycles will clarify whether this continues, stabilises, or accelerates.

Day 2 Re-test — Full 3 Samples

Sample	Initial (g)	Final (g)	Δm (g)	Loss (kg/m ² /hr)
Day 2 S1	12.5686	11.2211	1.3475	1.5488
Day 2 S2	12.6369	11.2767	1.3602	1.5720
Day 2 S3	13.1214	11.7309	1.3905	1.6170
Day 2 Avg	12.7756	—	1.3661	1.5793
vs. Original			−12.7%	
vs. Day 1			+1.4% — effectively stable	

Observations:

- R^2 values: S1 = 0.9994, S2 = 0.9990, S3 = 0.9987 — all excellent.
- Rate stable from Day 1 to Day 2: +1.4%, within normal variability.
- Δm virtually identical to Day 1 (1.347 g vs. 1.366 g), confirming water transport has not changed between cycles.

Day 3 Re-test — S1 & S2

Sample	Initial (g)	Final (g)	Δm (g)	Loss (kg/m ² /hr)
Day 3 S1	12.5636	11.2098	1.3538	1.5519
Day 3 S2	12.0199	10.7172	1.3027	1.5031
Day 3 S3	12.5686	11.2211	1.3475	1.5488
Day 3 Avg	12.3840	—	1.3347	1.5346
vs. Original			−15.2%	
vs. Day 2			−2.8% — stable	

Observations:

- R^2 values: S1 = 0.9995, S2 = 0.9994, S3 = 0.9994 — all excellent and consistent with all previous runs.
- **Rate remains stable:** 1.5793 (Day 2) $\rightarrow$ 1.5346 (Day 3), a change of only −2.8%. Combined with Day 1 to Day 2 (+1.4%), the plateau is confirmed across three consecutive re-test cycles.
- S2 sits slightly lower at 1.5031 with a notably lower initial weight (12.02 g vs. 12.56–12.57 g for S1 and S3), suggesting S2 absorbed less water during soaking on Day 3 — consistent with the initial weight variability observed across all reliability runs.
- Δm (avg 1.335 g) remains in the same range as Day 1 (1.347 g) and Day 2 (1.366 g), confirming stable water transport at the surface.

Reliability Summary — All Days

Day	n	Avg w_0 (g)	Avg Δm (g)	Avg Loss (kg/m ² /hr)	vs. Orig	vs. Prev
Original (Mar 18)	3	12.9669	1.5585	1.8091	—	—
Day 1 re-test	3	12.6129	1.3471	1.5575	−13.9%	−13.9%
Day 2 re-test	3	12.7756	1.3661	1.5793	−12.7%	+1.4%
Day 3 re-test	3	12.3840	1.3347	1.5346	−15.2%	−2.8%

Key Findings:

- **The plateau is confirmed across all three re-test cycles.** Day 1: 1.5575, Day 2: 1.5793, Day 3: 1.5346 — all within 2.8% of each other and all overlapping within $\pm 1\sigma$ error bands.
- The large drop occurred once, between the original run and Day 1 (−13.9%). No further systematic decline has been observed across Days 1–3.
- **Δm has stabilised:** 1.347 / 1.366 / 1.335 g across Days 1–3 — essentially flat, confirming water transport to the surface is not degrading with repeated use.
- **Conclusion:** the foam exhibits a one-time settling loss of ~ 14 –15% after first use, followed by stable performance across all subsequent cycles tested. This is a promising result for reusability.
- **Next steps:** continue repeat cycles beyond Day 3 to confirm the plateau holds long-term; run under matched ambient conditions to fully separate environmental effects from foam degradation.

Combined View — All Days and Samples

Observations:

- The combined weight trace panel makes the day-to-day shift immediately visible — the original run (solid blue) sits clearly above all re-test days, while Days 1, 2, and 3 overlap closely with each other.
- The shaded min-max bands show that within each day, sample-to-sample variability is small relative to the gap between the original and re-test days.
- The rolling rate panel confirms the simulator warm-up effect (upward rate drift) is consistent across all days and is not changing with repeated use cycles.
- The longevity trend (bottom right) most clearly shows the step-change after the first cycle and the subsequent plateau — the error bars from Days 1, 2, and 3 all overlap, providing strong evidence that foam performance has stabilised at its new operating level.

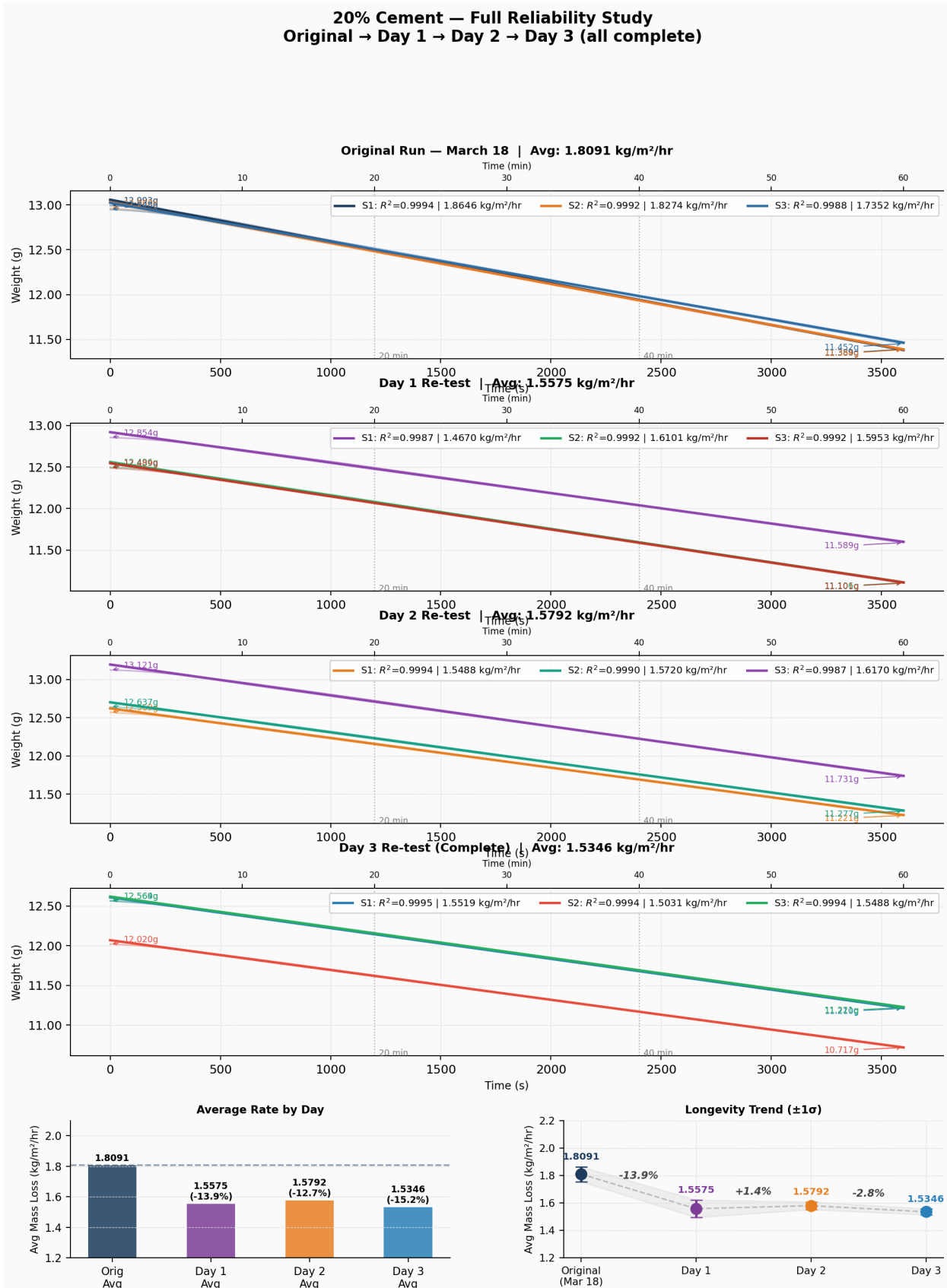


Figure 1: *

Top: original run (S1, S2, S3). Middle: reliability batch (S1, S2, S3) re-tested on the same specimens. Bottom left: bar comparison. Bottom right: longevity trend with $\pm 1\sigma$ error bands.

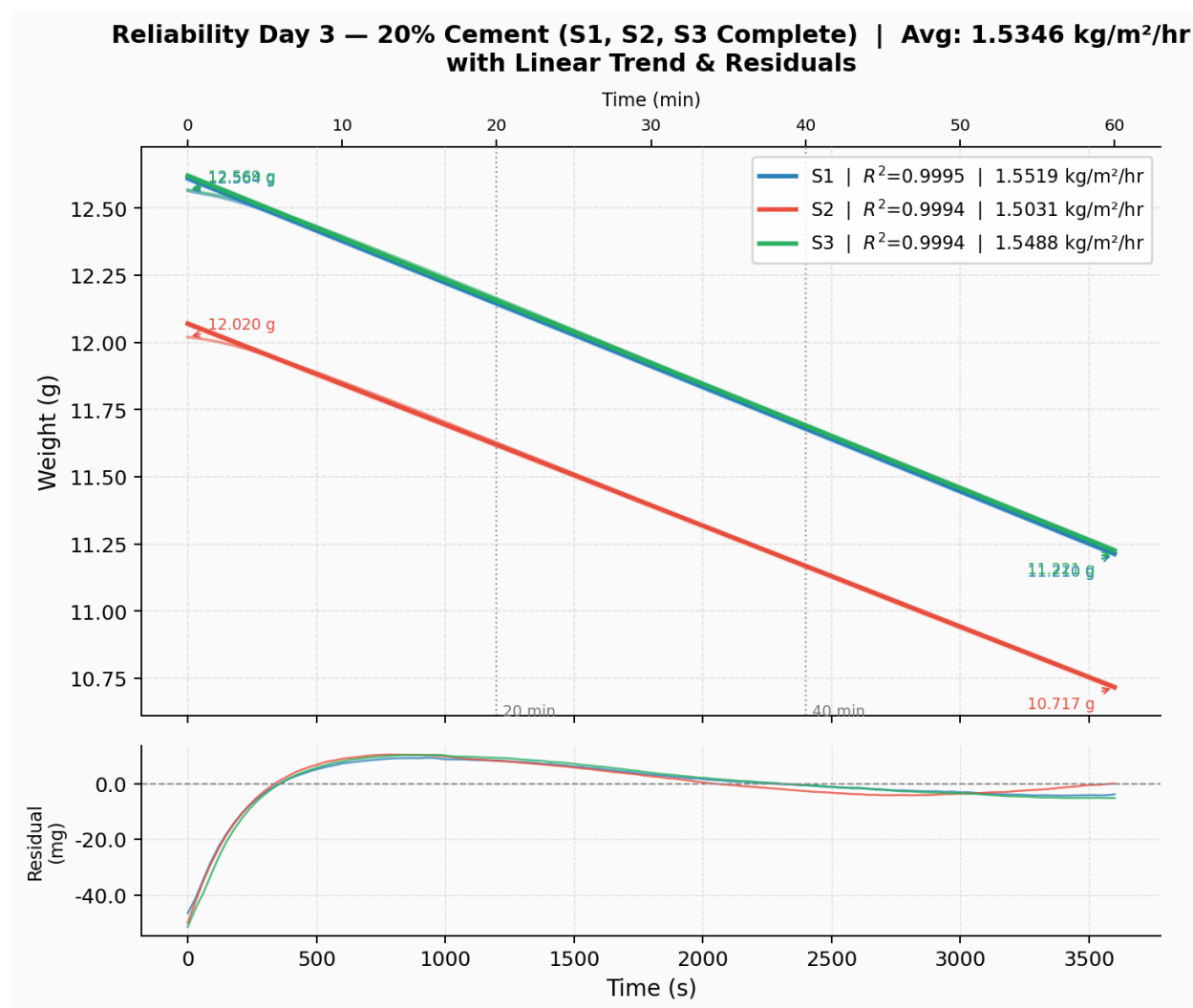


Figure 2: *
Day 3 reliability run — all three specimens (S1, S2, S3) complete

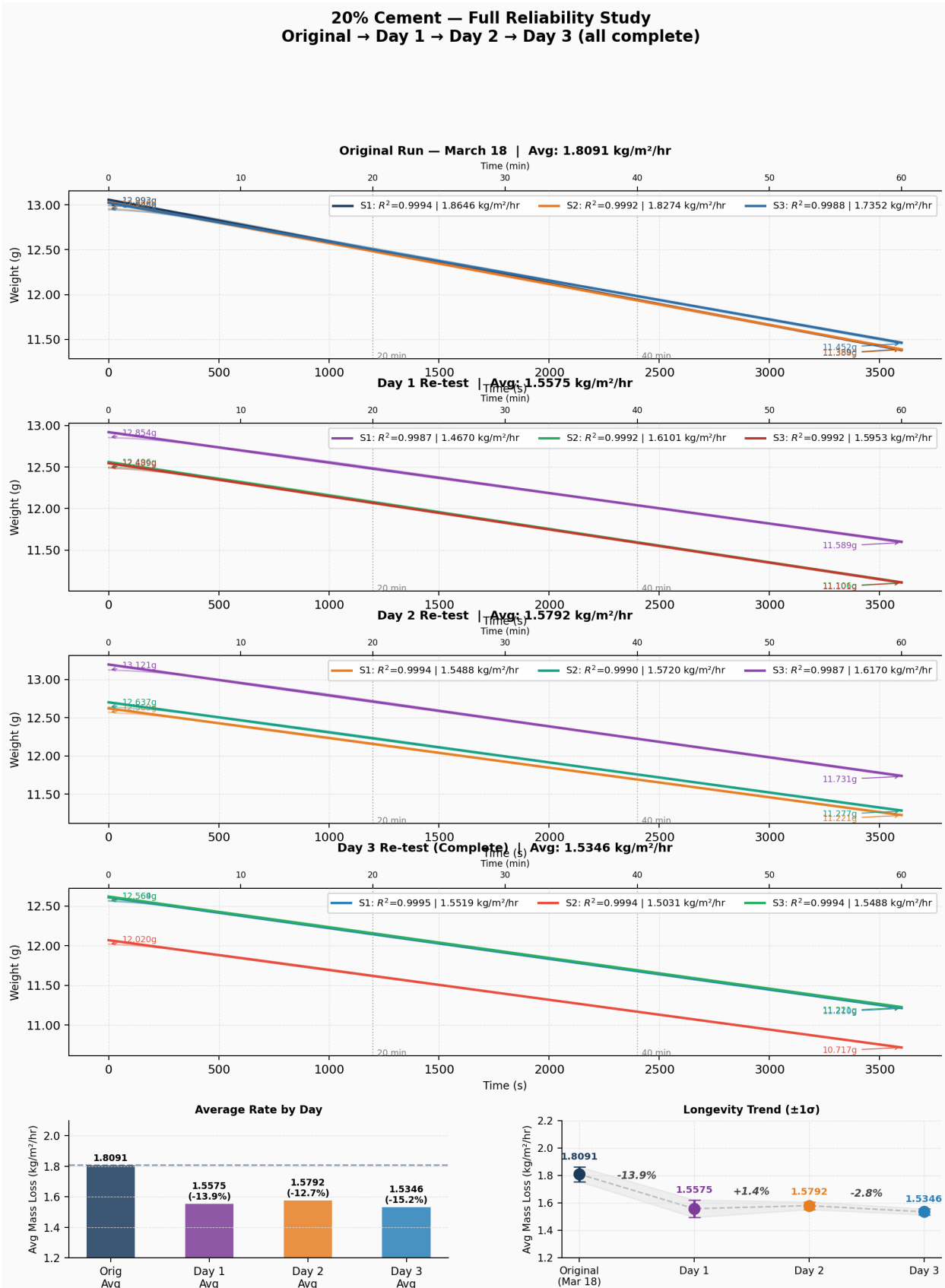


Figure 3: *

Original, Day 1, Day 2, and Day 3 re-tests stacked row by row. Bottom left: average rate by day. Bottom right: longevity trend with $\pm 1\sigma$ error bands.

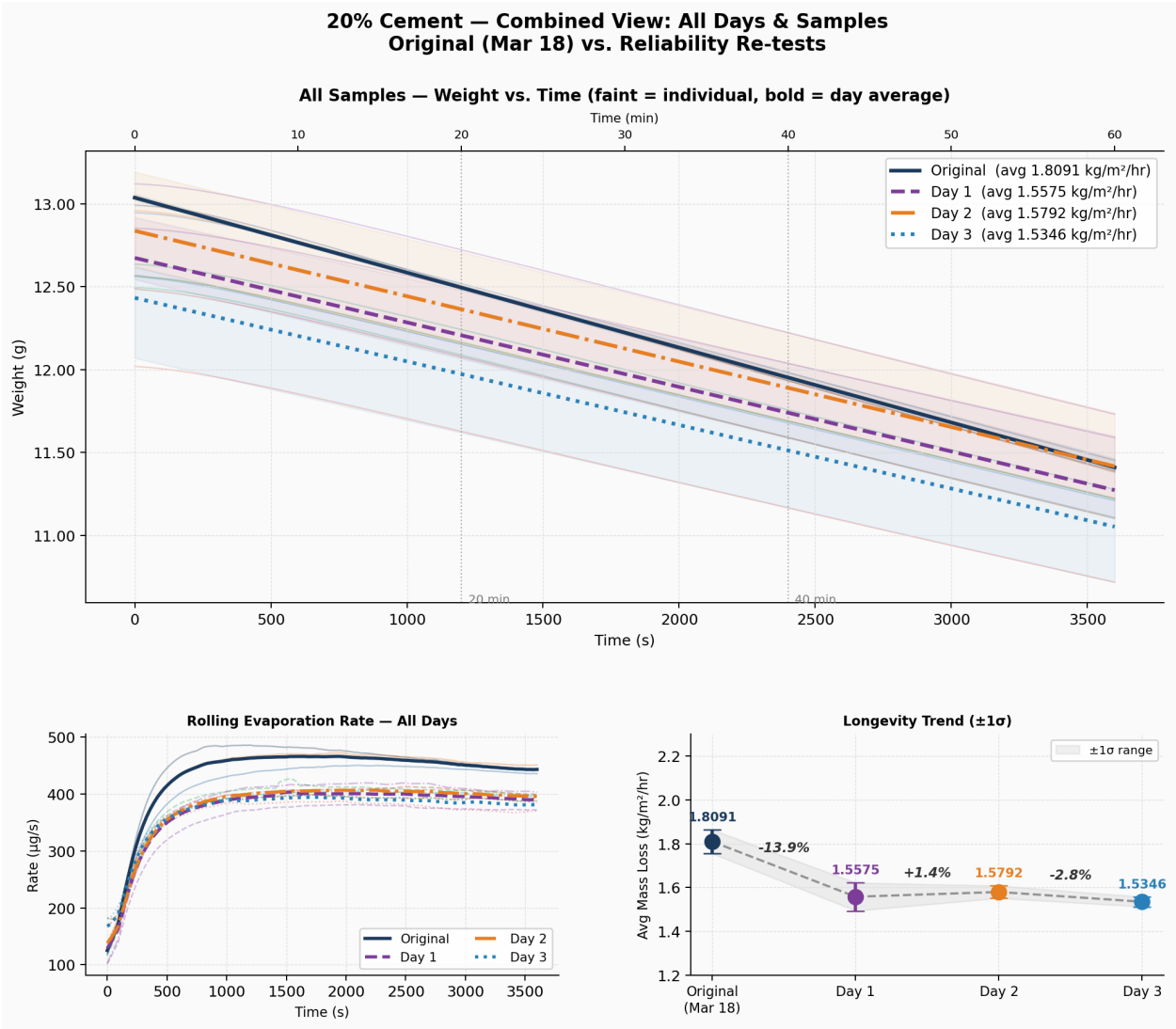


Figure 4: *

Top: all individual weight traces (faint) with bold day-average lines and shaded min-max bands. Bottom left: rolling evaporation rate for all days. Bottom right: longevity trend with $\pm 1\sigma$ error bands.